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MIXED MARKOV LATENT CLASS MODELS

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The process of change in a discrete characteristic is seldom the same for every respondent in a panel survey. Mixed Markov latent class models describe cross-tables of consecutive measurements in terms of several Markov chains. Each of these chains may be a stayer chain or some kind of mover chain. Markov chains may or may not be latent; i.e., they may or may not be unreliably measured. In this paper, we extend the mixed Markov latent class model to several subpopulations. We give formulas for estimation under equality restrictions when some parameters are fixed, and we give estimation priorities for several types of restrictions. Equality restrictions on parameters across subpopulations can be tested by means of a likelihood ratio test. To illustrate, we reanalyze a data set from Wiggins (1973).

1. INTRODUCTION

Since Lazarsfeld's (1978) overview of panel analysis, there has been a growing interest in methods for Markovian analysis of change in discrete data. Models for the analysis of event history

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data (Tuma and Hannan 1984) and a great variety of latent class models for the analysis of discrete time panel data, also considered as (latent) mixtures of Markov chains (Poulsen 1982; Langeheine and Van de Pol 1990), have been developed. In this paper, we focus on discrete time mixed Markov latent class (MMLC) models. These models make statements about transitions from one point in time to another, i.e., from one panel occasion to another. They do not describe the process of change between occasions, as continuous time models do. Other models for the analysis of change in discrete variables are treated by Clogg, Eliason, and Grego (1989), Duncan (1985), and Plewis (1985).

Although a nice description of multiwave panel data can be obtained with MMLC models, there are reasons to extend them to simultaneous analysis of several subpopulations. The assumption of homogeneity of subpopulations in the latent classes should often be relaxed. If the population is split up according to a characteristic that is relevant for the transition probabilities or for other parameters, the parameters that are allowed to vary across subpopulations will be estimated with less specification bias but, in most cases, with a larger standard error. Moreover, insight into the heterogeneity of the process increases. Mobility in the labor market, for instance, is quite different for younger and older people.

In section 2, we review MMLC models by briefly treating familiar special cases, such as the single Markov chain, the mover-stayer model, the black-and-white model, and the latent Markov model. In section 3, we examine other special cases, such as a mixed Markov model with transitions between chains and a multivariate MMLC model. We also show how the MMLC framework can handle higher-order Markov chains. In section 4, we extend the MMLC models. We describe how simultaneous analysis of several populations is performed and give formulas for estimation when some parameters are fixed or set equal to other parameters. The ability to set any parameter equal to any other parameter, together with the ability to fix parameters, greatly enlarges the range of feasible MMLC models. In section 5, we apply this approach to a specific data set.

2. MIXED MARKOV LATENT CLASS MODELS: FAMILIAR SPECIAL CASES

MMLC models can be used to analyze several (T) consecutive measurements, $x^1, x^2, \ldots, x^T$, with realizations $i, j, \ldots, m$, of the same polytomous variable. They can be used to analyze attitudes, vote intentions, employment status, or the recent occurrence of some behavior, such as buying. MMLC models can be applied to panel data and to data obtained from a survey with retrospective questions or from administrative files. Analysis with MMLC models focuses on the types of change that occur.

An extreme case of MMLC models is a model that ascribes all observed change to measurement error, response uncertainty, and the like. Such a no-latent-change model is the latent class model (Lazarsfeld and Henry 1968; Dayton and Macready 1983; Langeheine 1988; Hagenaars 1988). It assumes that each respondent belongs to one latent class (a) of a limited number (A) of latent classes, variable y. Nevertheless, change can be observed, because respondents have a probability lower than 1 of giving response i ($i = 1, \ldots, J$). Every respondent in class a has the same response probability, $\rho^t_{i|a}$, of answer i at occasion t, with $\Sigma_i \; \rho^t_{i|a} = 1$. (Superscripts indicate occasions.) We will give formulas for three occasions. Extension to more occasions is straightforward.

The proportion of the population that belongs to class a is denoted by δ_a. A respondent's answers on several items are assumed to be independent, conditional on the latent class that he or she belongs to. Hence, the probability that a respondent from class a will respond i, j, and k on three consecutive occasions is $\delta_a \; \rho^1_{i|a} \; \rho^2_{j|a} \; \rho^3_{k|a}$. Summing over the latent classes, we obtain the proportion that will answer i, j, and k:

$$P_{ijk} = \sum_{a=1}^{A} \delta_a \, \rho^1_{i|a} \quad \rho^2_{j|a} \quad \rho^3_{k|a} \quad . \tag{1}$$

In a (panel) sample an estimate, p_{ijk}, is found, from which the parameters can be estimated. In this case of consecutive measurements, the latent class model says that every manifest variable is an imperfect measurement of the same static underlying factor (y). The response probabilities, $\rho^t_{i|a}$, are the typical characteristics of a set of classes and an item, like a factor loading in a principal components analysis. For panel data the number of latent classes, A, is usually assumed to be equal to the number of manifest response categories, J. Thus, reliability is high if response probability $\rho^t_{i|a}$, with $a = i$, is high. Moreover time-homogeneity of response probabilities can be assumed:

$$\rho^1_{i|a} = \rho^2_{i|a} = \rho^3_{i|a} . \qquad (2)$$

In case of time-homogeneity, equation (1) can be written without superscripts: $P_{ijk} = \Sigma_a \delta_a \rho_{i|a} \rho_{j|a} \rho_{k|a}$.

Another basic MMLC model is the single Markov chain model (M). It assumes that only the state on the most recent occasion is important for predicting the present state. Moreover, it assumes that the population is homogeneous, i.e., that everybody who is in state j at occasion 2 has the same transition probability, $\tau^{3\,2^*}_{k|j}$, of being in state k at occasion 3 (regardless of their state at occasion 1):

$$P_{ijk} = \delta^{1^*}_i \tau^{2\,1^*}_{j|i} \tau^{3\,2^*}_{k|j} , \qquad (M) \quad (3)$$

$\Sigma_j \tau^{2\,1^*}_{j|i} = 1$, $\Sigma_k \tau^{3\,2^*}_{k|j} = 1$. The δ^{1^*} in this equation refer to the manifest proportions (at occasion 1), whereas the δ in equation (1) refer to the latent proportions (valid at any occasion). We note, in passing, that being in the same state on two occasions is no guarantee that no change has occurred in the intermediate period. Continuous time models aim at incorporating these changes too (by making an assumption about the change in the rate of change); discrete time models, like MMLC models, don't.

These two models, the latent class model and the single Markov chain, are the constituting elements of MMLC models. Population heterogeneity can be modeled by assuming more than one Markov chain, and unreliability can be modeled by defining Markov chains as latent. Before dealing with these more complicated models, we first discuss some useful special cases of the single Markov chain: independence, stationarity, and no change.

Independence. If someone's current state is independent of his state at the previous occasion, he may be thought of as flipping a biased coin, possibly another coin at each occasion (Converse 1964; Duncan, Stenbeck, and Brody 1988). For more than two categories, a die would be a more appropriate random generator. In terms of the probability of transition from state i at time t to state j at time $t + 1$, $\tau_{j|i}^{t+1t^*}$, this independence chain (Ip) is restricted by

$$\tau_{j|1}^{t+1t^*} = \tau_{j|2}^{t+1t^*} = \ldots \tau_{j|J}^{t+1t^*} . \qquad \text{(Ip)} \quad (4)$$

When transition probabilities, $\tau_{j|i}^{t+1t^*}$, are arranged in a matrix, with i as row index and j as column index, all rows are the same and are equal to the marginal distribution at time $t + 1$, though they may differ for each transition period.

Stationarity. Any Markov chain can be restricted to be stationary, $P_{ijk} = \delta_i^{1^*} \tau_{j|i}^{*} \tau_{k|j}^{*}$, which means that transition probabilities are homogeneous in time:

$$\tau_{j|i}^{2\,1^*} = \tau_{j|i}^{3\,2^*} = \ldots . \qquad (5)$$

If an independence chain (equation (4)) is stationary, then the probability of each state, the marginal distribution, is the same at each occasion ($t > 1$); i.e., the parameters in equation (4) are the same for every transition period. If a time-homogeneous marginal distribution is assumed, starting at occasion 1, then these transition probabilities are also equal to the marginal distribution at occasion 1, $\delta_j^{1*} = \tau^*_{j|i}$. In the coin flipping metaphor, this means that at each occasion the respondent is flipping the same biased coin. Converse (1964) proposed a model that incorporates individuals who respond not only independently at consecutive occasions (Ip) but even randomly (model Rr); i.e., they flip an unbiased coin, $\delta_j^{1^*} = \tau^*_{j|i} = 1/J$, at every occasion.

No change. The no-change case can be modeled by assuming that the same categories apply for all occasions. The no-latent-change case is modeled this way in equation (1), where the proportion in latent class a, δ_a, has no subscript for i (occasion 1), j (occasion 2), or k (occasion 3). However, a more uniform notation is obtained if a Markov chain (equation (3)) is used with probability 1 of staying in the same state as before (model S):

$$\begin{aligned} \tau_{j|i}^{*} &= 1 \quad \text{for } i = j, \\ \tau_{j|i}^{*} &= 0 \quad \text{otherwise.} \end{aligned} \qquad \text{(S)} \quad (6)$$

Arranged in a matrix, this no-change or stayer transition matrix is the identity matrix.

In the social sciences the assumption of homogeneous change is usually a gross simplification. This assumption is relaxed somewhat in the black-and-white model (Converse 1964; Taylor 1983; Brody 1986; Langeheine and Van de Pol 1989) and the mover-stayer model (Blumen, Kogan, and McCarthy 1966). Poulsen (1982) formulated the more general mixed Markov model, which assumes S latent classes (chains), each having their own transition probabilities:

$$P_{ijk} = \sum_{s=1}^{S} \pi_s \, \delta^{1*}_{i|s} \quad \tau^{*}_{j|is} \quad \tau^{*}_{k|js} . \qquad \text{(MM)} \quad (7)$$

Also, the proportion in state i at occasion 1, $\delta^{1*}_{i|s}$, may be different for each chain. The proportion of the population that behaves according to Markov chain s is denoted by π_s.[1] Of course a nonstationary version, with superscripts for every transition period (as in equation (3)), can also be conceived. Davies and Crouchley (1986) used a mixed Markov model with restricted $\delta^{1*}_{i|s}$ parameters.[2] All mixed Markov models restrict membership in chain s to be permanent; i.e., there is no turnover between chains.

The black-and-white model and the mover-stayer (MS) model can be viewed as mixed Markov models with two chains and additional restrictions. The mover-stayer model has one unrestricted chain (the mover chain) and one chain restricted according to equation (6) (the stayer chain). The black-and-white model is a mover-stayer model with a mover chain that displays only independent responses (IpS, equations (4) and (6)) or even random response (RrS). Duncan et al. (1988) showed that for a dichotomy observed on three occasions, the nonstationary IpS model has the same fit as the Rasch model with three items. If, however, more occasions are available, the IpS and Rasch models are not the same.

Of course other, new, restricted two-chain mixed Markov models can be conceived, such as a combination of movers (equation

[1] For event history data, Heckman and Singer (1984) also modeled heterogeneity with a limited number of latent classes (i.e., without distribution assumptions), in this case in continuous time.

[2] In their reanalysis of the Morgan, Aneshensel, and Clark (1983) data, Davies and Crouchley assumed a stationary process with steady state marginal distributions at occasion 1, i.e., for a dichotomy, $\delta^{1*}_{1|s} = \tau^{*}_{2|1s}/(\tau^{*}_{1|2s} + \tau^{*}_{2|1s})$ and $\delta^{1*}_{2|s} = \tau^{*}_{1|2s}/(\tau^{*}_{1|2s} + \tau^{*}_{2|1s})$. They write ξ_{pr} for $\tau^{*}_{1|1s}$ $(s=r)$ and ξ_{qr} for $\tau^{*}_{2|2s}$. Moreover, π_s is denoted as ρ_r.

(3)) and people whose responses are independent of previous responses (equation (4)) (a mover-independence model) or a mixture of movers and people who respond randomly (a mover-random-response model).

The latent Markov (LM) model (Wiggins 1973) is a combination of the unrestricted latent class model and a single Markov chain. According to this model, the whole population changes according to one Markov chain, but measurement error causes the observed cross-table to diverge from what would be observed if reliability were perfect (model M, equation (3)):

$$P_{ijk} = \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \delta_a^1 \, \rho_{i|a}^1 \quad \tau_{b|a} \quad \rho_{j|b}^2 \quad \tau_{c|b} \quad \rho_{k|c}^3 . \qquad \text{(LM)} \quad (8)$$

Of course, the stationarity assumption (5), which is implicit in equation (8), can be relaxed. The response probabilities of the first and the last occasion cannot be estimated from observed data for the latent Markov model without further restrictions. It is often reasonable to assume that they are equal to those of the nearest occasion, $\rho_{i|a}^1 = \rho_{i|a}^2$ and $\rho_{k|c}^3 = \rho_{k|c}^2$, giving equation (8) for three occasions. For four or more occasions, one does not have to assume that response probabilities are time-homogeneous. Furthermore, restricting the transition probabilities, τ's, according to equation (6) reduces the latent Markov model (LM) to the latent stayer model (LS), which was given in equation (1) without transition probabilities as the latent class model (L). A latent independence model with one latent class—a model with more classes is not identified—is just another formulation of the manifest independence model (equation (4)).

The latent Markov model of equation (8) can be extended to a mixture of several latent Markov chains (Poulsen 1982; Langeheine and Van de Pol 1990):

$$P_{ijk} = \sum_{s=1}^{S} \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \pi_s \, \delta_{a|s}^1 \quad \rho_{i|as}^1 \quad \tau_{b|as} \quad \rho_{j|bs}^2 \quad \tau_{c|bs} \quad \rho_{k|cs}^3 . \qquad \text{(LMM)} \quad (9)$$

This model can also be seen as a mixed Markov model that is measured unreliably. The mixed Markov model is a special case of equation (9), which occurs if the number of latent classes is equal to the number of manifest categories ($A = J$, $B = J$, etc.) and reliability is perfect ($\rho_{i|as} = 1$ for $a = i$, $\rho_{i|as} = 0$ otherwise). All previously mentioned models are special cases of equation (9). To emphasize that these models are all latent class models, we refer to them as mixed Markov latent class models (MMLC), using latent mixed Markov as a name for a latent Markov model that is restricted to be mixed Markov on a latent level. Restrictions should be applied to make the latent mixed Markov model identified, for instance, by restructuring it to a latent mover-stayer model with perfect reliability for the stayers only (Langeheine and Van de Pol 1990).

Once a cross-table is observed in a sample, maximum likelihood (ML) estimates can be obtained using direct estimators (Anderson 1954; Anderson and Goodman 1957), the EM algorithm (Poulsen 1982; Van de Pol and De Leeuw 1986; Fuchs and Greenhouse 1988; Van de Pol and Langeheine 1989; Abdel-Aty 1989), or other algorithms (Morgan et al. 1983; Bye and Schechter 1986; Davies and Crouchley 1986). The fit of nested models can be compared with a likelihood ratio (LR) test. Because no easy-to-use program for latent class models with mixed Markov-type restrictions was available, we developed PANMARK (Van de Pol, Langeheine, and de Jong 1989).[3] Mixed Markov models (except the two-chain mover-stayer model and submodels) should be estimated with several sets of starting values, because these models may have local maxima.

This approach utilizes all available data (full-information ML). An LR test of, for instance, the hypothesis that a Markov chain is stationary can be obtained by simply comparing the fit of

[3] PANMARK is a user-friendly program written in Turbo Pascal for mixed Markov and latent Markov models and simultaneous analysis of several subpopulations. PANMARK has many options: several standard models and standard restrictions, input of a restrictions file (any set of parameters can be fixed or restricted to be equal to any other set of parameters), standard deviations, correlations between parameters, first-order derivatives of the log-likelihood with respect to each independent parameter, and an identification test. The PANMARK program and a user's manual can be purchased from the Netherlands Central Bureau of Statistics, Department of Automation, Statistical Informatics Unit, P.O. Box 959, 2270 AZ Voorburg, The Netherlands, for about \$47 plus about \$13 handling costs. The program is available on a $5\frac{1}{4}$ inch diskette for use with DOS on the PC, XT, or AT.

a nonstationary Markov chain with the fit of a stationary Markov chain. Limited-information methods are appropriate for testing stationarity (Anderson and Goodman 1957; Bishop, Fienberg, and Holland 1975; Markus 1979) but not for testing the fit of the single Markov chain model. To test the fit of a nonstationary Markov chain, Bishop et al. used a loglinear model. The MMLC approach is much simpler.

3. OTHER SPECIAL CASES

The mixed Markov model says that people belong permanently to one of S latent chains on all occasions. In many cases, however, this assumption of no turnover between chains is unrealistic. For example, an individual who belongs to a chain of people who are usually employed will shift, upon retirement, into a chain of people who are economically inactive.

To incorporate turnover between chains into the mixed Markov model, we first formulate it as a latent Markov model. For a dichotomous characteristic, we can formulate a two-chain mixed Markov model as a latent Markov model with $A = 4$ $(S \times J)$ latent classes and perfectly reliable measurements (see Table 1). Moreover, if turnover between chains does not exist, some transition probabilities should be zero.

The chain 1 transition probabilities are in the upper left block of the right side of Table 1, and the chain 2 transition probabilities are in the lower right block. Once the mixed Markov model is formulated in this way, it is easily generalized to a model with turnover between all latent classes. The zero restrictions on the transitions in Table 1 should be removed, at least partially. Response probabilities still have to be fixed at 0 and 1. An application of this approach to five-wave job mobility data is in preparation.

Higher-order Markov chains and multivariate models like cross-lagged panel models can also be incorporated in the MMLC framework. Higher-order Markov chains relax the Markovian assumption of a process without memory, i.e., that knowledge of someone's state at occasion 2 is sufficient to predict someone's state at occasion 3 (equation (3)). In a second-order Markov model,

TABLE 1
Structural Zeroes in Response Probabilities, $\rho_{i|a}$, and Transition Probabilities, $\tau_{b|a}$, of the Mixed Markov Model, Formulated as a Latent Markov Model

	Response Probabilities		Transition Probabilities			
			$b = 1$	$b = 2$	$b = 3$	$b = 4$
	$i = 1$	$i = 2$	$j = 1$	$j = 2$	$j = 1$	$j = 2$
$a = 1$, chain 1, category 1 ($i = 1$)	1	0	$\tau_{1\|1}$	$\tau_{2\|1}$	0	0
$a = 2$, chain 1, category 2 ($i = 2$)	0	1	$\tau_{1\|2}$	$\tau_{2\|2}$	0	0
$a = 3$, chain 2, category 1 ($i = 1$)	1	0	0	0	$\tau_{3\|3}$	$\tau_{4\|3}$
$a = 4$, chain 2, category 2 ($i = 2$)	0	1	0	0	$\tau_{3\|4}$	$\tau_{4\|4}$

transition probabilities to occasion 3 depend on the state at occasion 2 and on the state at occasion 1. Second-order Markov chains are not special cases of MMLC models, but they can be analyzed within the MMLC framework by manipulating the cross-table of x^1, x^2, . . ., x^T (Poulsen 1982). A cross-table of xx^1, xx^2, . . ., xx^{T-1} (with realizations f, g, . . .), where xx^1 is a variable with all possible combinations of x^1 and x^2 as categories, etc. should be analyzed. The xx cross-table will have structural zeroes, as will the matrices with transition probabilities (see Appendix A).

The trick of analyzing consecutive measurements of a cross-table instead of consecutive measurements of one variable can more naturally be used to analyze the transition probabilities of more than one variable simultaneously—i.e., the 16-fold table (Duncan 1985)—and to fit MMLC models on these data. In this multivariate case, the data will not contain structural zeroes.

4. MODELS FOR SEVERAL SUBPOPULATIONS

4.1. *The Extended Model*

Using the MMLC models of section 2, we can describe T-way tables of T consecutive measurements of one target variable. Most of the cited references give examples of one or more specific MMLC models. Seeing these examples, however, one may ask, What kind of people are in those latent chains and classes? To answer this question, we can fit MMLC models for subpopulations, such as younger people and older people or low-education people and high-education people.

However, sample sizes for subpopulations will be smaller; therefore, models with less parameters will have to be fitted. More realistic, more complex models could be fitted if some parameters turned out to be equal across subpopulations. Also, for mixed Markov models, a simplification may result if one sort of turnover is not present in all subpopulations. For example, if we analyze two subpopulations—one composed of people who wanted to move at the beginning of an observation period and one composed of people who did not want to move—a stayer chain (or a chain with little turnover) would probably be in the model only for those who did not want to move.

To enable this sort of analysis, the latent mixed Markov model is extended with a parameter for the proportion, γ_h, that belongs to subpopulation h ($h = 1, \ldots, H$), and all other parameters are considered conditional on subpopulation h; i.e., all (or some) parameters may be different in each subpopulation:

$$P_{hijk} = \gamma_h \sum_{s=1}^{S} \sum_{a=1}^{A} \sum_{b=1}^{B} \sum_{c=1}^{C} \pi_{s|h}\, \delta^1_{a|sh}\, \rho^1_{i|ash}\, \tau_{b|ash}\, \rho^2_{j|bsh}\, \tau_{c|bsh}\, \rho^3_{k|csh}\,. \quad (10)$$

Analogous extensions to simultaneous analysis of several subpopulations were made by Clogg and Goodman (1984, 1985, 1986) for unrestricted latent class analysis and by Jöreskog (1971) for factor analysis.

Not only models for panel data, the latent mixed Markov, latent Markov, mixed Markov, mover-stayer and black-and-white models, and similar alternatives, are MMLC models. Cross-sectional latent structure analysis on a single multidimensional contingency table, or on several simultaneously (Clogg and Goodman 1984, 1985, 1986), can be performed by leaving transition parameters out of the latent Markov model (equation (1) or the latent stayer model) and by using time points as indicator variables. The special case of simultaneous analysis of one variable at one point in time (Van der Heijden, Mooijaart, and De Leeuw 1989; De Leeuw, Van der Heijden, and Verboon 1990) can be viewed as a latent stayer model, with γ_h, $\delta_{a|h}$, and $\rho^1_{i|ah}$ as parameters, and as a mixed Markov model, with γ_h, $\pi_{s|h}$, and $\delta^{1*}_{i|sh}$ as parameters.

4.2. *Estimation Under Restrictions*

ML estimates for the proportion of each subpopulation, γ_h, can be obtained separately from the other parameters. In a random sample, the parameter $\hat{\gamma}_h$ is simply the proportion of subpopulation h in the sample. The other parameters can be estimated with the EM algorithm (Dempster, Laird, and Rubin 1977), which is an iterative process of E steps and M steps. In the E step, the expectation of the full table of subpopulations × latent classes × observed response patterns, with cell proportions $\hat{\theta}_{hsabcijk}$, is computed on the basis of the parameter estimates from the last iteration and the observed data, P_{hijk}. In the M step, auxiliary ML parameter estimates are obtained for the latest full table.

The EM algorithm for the mixed Markov model (equation (7)) was described by Poulsen (1982) and Van de Pol and Langeheine (1989); the EM algorithm for the latent Markov model (equation (8)) was described by Poulsen (1982) and Van de Pol and De Leeuw (1986). Both are implemented in PANMARK. Abdel-Aty (1989) gives an EM algorithm to estimate the stationary latent Markov model from bivariate tables of consecutive measurements.[4] For the extension to simultaneous estimation in several groups, only general rules for estimation under equality constraints have to be found. Moreover, fixing parameters to their starting values will be considered.

An auxiliary ML estimate of a parameter in, for instance, the row with index s, conditional on h, $\pi_{s|h}$, is the ratio of the following two quantities. The numerator is found by summing the full table over all indices, except s and h, $\hat{\theta}_{hs++++++}$, and the denominator (the row total) is found by summing over all indices, except those that were conditioned upon, in this case h, $\hat{\theta}_{h+++++++}$. To simplify notation, we will write the numerator elements, $\hat{\theta}_{hs++++++}$, in a matrix with elements $\hat{\theta}^{\pi}_{de}$, with row index d instead of h and element index e instead of s.

In fact, each MMLC parameter can be viewed as a proportion, $\varphi_{e|d}$, in a row d with parameters that sum to 1. For each combination of s and h there is a row with parameters $\hat{\delta}_{a|sh}$ and row sum $\hat{\delta}_{+|sh} = 1$. The numerators for $\hat{\delta}_{a|sh}$, $\hat{\theta}_{hsa+++++}$, will again be written in a matrix, with elements $\hat{\theta}^{\delta}_{de}$, d taking all combinations of values of h and s. Both matrices can be stacked, one below the other, in a two-way table with elements $\hat{\theta}_{de}$. Short rows are filled out with zero cells, if necessary. Also, rows for the numerators of other parameters, ρ's and τ's, will be added. The elements $\hat{\theta}_{de}$ in

[4] The same estimates can be found by treating each bivariate table as if it represented one of several subpopulations and by assuming that response probabilities and transition probabilities are equal across subpopulations (i.e., bivariate tables).

this two-way table can be assumed to be multinomially distributed, as are the cells $\hat{\theta}_{hsabcijk}$ in the multiway table. Thus, the problem of estimating π, δ, ρ, and τ parameters under restrictions is reduced to the problem of estimating only one sort of parameter under restrictions: conditional proportions for a large two-way table. In several ways this approach is more general than the approach of Goodman (1974). Here, several types of parameters can be set equal, even in the presence of fixed parameters. Moreover, estimation priority for several types of restrictions will be given. Goodman, on the other hand, also considered equality restrictions on sums of parameters, which we do not do here.

Two extra subscripts, f and q, are added to a numerator, $\hat{\theta}_{defq}$, to indicate whether the corresponding parameter, $\varphi_{e|dfq}$, is assumed fixed or equal to other parameters. If subscript $f = 1$, the parameter is fixed to $\varphi_{e|d10}$; if $f = 0$, the parameter is not fixed. All parameters with $q = 0$ are not set equal. All parameters with $q = 1$ make up the first set of equal parameters, those with $q = 2$ the second set, etc. Fixed parameters are not set equal to other parameters. A bar above an index indicates one realization of that index instead of any possible value.

If no parameter in row $\bar{d}$ is fixed or set equal to other parameters ($f = 0$ and $q = 0$ for all e), the (auxiliary) ML estimate of element $\bar{e}$ in row $\bar{d}$, $\hat{\varphi}_{e|\bar{d}00}$, is the ratio of element $\hat{\theta}_{\bar{d}\bar{e}00}$ and its row total, $\hat{\psi}_{d\bar{e}00}$ (with subscripts as $\hat{\theta}_{\bar{d}\bar{e}00}$):

$$\hat{\varphi}_{\bar{e}|\bar{d}00} = \frac{\hat{\theta}_{\bar{d}\bar{e}00}}{\hat{\psi}_{\bar{d}\bar{e}00}}, \tag{11}$$

where ψ denotes a sum of θ's over e. A row sum ψ_{de} is viewed as an attribute of a row d and as an attribute of an element e in that row.

If one or more parameters are fixed in a row d, the row sum is adapted to this. Let $\psi_{\bar{d}e00}$ be the sum (over e) of elements $\theta_{\bar{d}e00}$ of free parameters in row $\bar{d}$, and let $\varphi_{+|\bar{d}10}$ be the row sum of fixed parameters. The ML estimate of free element $\bar{e}$ in row $\bar{d}$ can be shown to be

$$\hat{\varphi}_{\bar{e}|\bar{d}00} = \frac{\hat{\theta}_{\bar{d}\bar{e}00}}{\hat{\psi}_{\bar{d}\bar{e}00}/[1 - \varphi_{+|\bar{d}10}]}. \tag{12}$$

In this way the sum of free and fixed parameters in row $\bar{d}$, $\varphi_{+|\bar{d}f0}$, will again be 1. Because each parameter row has one degree of freedom less than the number of free parameters in that row, one free parameter in each row must remain unrestricted (or the restriction on the last parameter must be consistent with the other parameter restrictions for that row).

If parameter $\varphi_{\bar{e}|\bar{d}0q}$ belongs to equality set $\bar{q}$, the ML estimate is found by summing the numerators and their denominators (row sums) for which $q = \bar{q}$ and then dividing that numerator sum by the denominator sum. Let $\hat{\eta}_{\bar{d}\bar{e}0\bar{q}}$ be the sum of all elements, $\hat{\theta}_{de0\bar{q}}$, with indices $\bar{d}$ and $\bar{e}$ that correspond to equality set $\bar{q}$, and let $\hat{\zeta}_{\bar{d}\bar{e}0\bar{q}}$ be the sum of all relevant row sums, $\hat{\psi}_{de0\bar{q}}/(1-\varphi_{+|d10})$, for set $\bar{q}$, taking care not to sum over elements that are fixed and correcting for fixed parameters, as in the denominator of the right-hand side of equation (12). (Note that if an equality restriction holds for two parameters within the same row $\bar{d}$, the same row sum $\hat{\psi}_{de0\bar{q}}/(1-\varphi_{+|d10})$ will be counted twice, because row sum $\hat{\psi}_{d10\bar{q}}/(1-\varphi_{+|d10})$ has the same value as row sum $\hat{\psi}_{d20\bar{q}}/(1-\varphi_{+|d10})$.) The estimate for parameter $\bar{e}$ in row $\bar{d}$ from equality set $\bar{q}$ is

$$\hat{\varphi}_{\bar{e}|\bar{d}0\bar{q}} = \frac{\hat{\eta}_{\bar{d}\bar{e}0\bar{q}}}{\hat{\zeta}_{\bar{d}\bar{e}0\bar{q}}}. \tag{13}$$

Equation (13) also holds if complete rows are set equal. This is the case, for instance, if transition probabilities or response probabilities are assumed time-homogeneous. However, equation (13) does not always apply if a parameter in row $\bar{d}$ belongs to a set of equal parameters that does not contain the same number of parameters as the other equality sets for that row. This is the case, for instance, if two rows are assumed equal and the additional restriction holds that parameters 2 and 4 of these rows are equal. Then the other parameters belong to shorter equality sets than elements 2 and 4. Parameters from the smallest equality sets, subequal parameters, cannot be estimated with equation (13). Parameters from the larger equality sets in a row, supraequal parameters, can be estimated with equation (13).

If only some parameters in row $\bar{d}$ are set equal to other parameters, these parameters should be estimated first. The same holds for parameters from supraequality sets. These estimates should be considered fixed for the estimation of the other parameters. Then free parameters can be estimated using equation (12), subequal parameters using equation (13).

This row-wise approach can handle almost all equality restrictions. Exceptions occur if a set of equal parameters has some members that have estimation priority in one row and other members that do not have priority in another row. For example, take two rows from a 4×4 transition table, where transitions are restricted in row 1 by 0 0 0 1 (free, free, free, member of equality set 1) and in row 2 by 2 2 2 1. In this table, set 1 has estimation priority in row 1, and set 2 has priority over set 1 in row 2. The parameter value of equality set 1 is determined as soon as the parameter value of equality set 2 is estimated (because the row sum is 1). Therefore, in such a case, the parameter value of the equality set with mixed priority (set 1 in this case) should be determined only on the basis of information on the row (row 2 in this case) where this set (1) has no priority.

Therefore, the final estimation priority is as follows. First, all parameters that are part of an equality set are estimated, except the so-called subequal parameters and the parameters with mixed priority (conditional on fixed parameters). Then, all parameters with mixed priority are estimated, considering all previously estimated parameters as fixed. Finally, all free and subequal parameters are estimated, except those with mixed priority, again considering all previously estimated parameters as fixed. In this way fixed, equal, and free parameters are handled in PANMARK.

5. AN EXAMPLE OF SIMULTANEOUS ANALYSIS IN TWO GROUPS

Wiggins (1973) analyzed several three-wave tables from a panel that emerged from a nationwide program called Education and Neighborhood Action for a Better Living Environment (ENABLE). The panel consisted of some 1,000 parents from about 61 low-income communities across the U.S. The parents engaged in group discussions primarily about problems of raising children. These group discussions, as well as other services provided by the ENABLE program, took place between the first and second waves

TABLE 2
The Fit of MMLC Models with Several Restrictions

Model	Time-Homogeneous Response Probabilities (ρ)	Stationarity, Transition Probabilities (τ^*, τ)	LR	*df*
Two-class latent class	+		84.9	4
Two-class latent class	−		0	0
First-order Markov		+	18.8	4
First-order Markov		−	14.4	2
Second-order Markov		−/+	0	0
Latent Markov	+	+	1.2	2
Latent Markov	+	−	0	0
Wiggins	+	+	2.9	4
Wiggins	+	−	2.1	3
Mover-stayer		+	7.4	2
Mover-stayer		−	0	0
Mover-independence		+[a]	0.4	2
Mover-independence		−/+[a]	0.0	0
Mover-random-response		+	0.5	3
Mover-random-response		−	0.3	1

Note: "+" = true, "−" = false. See text for explanation of "+/−."

[a]For the independence chain, time-homogeneity of marginal distributions is assumed, $\delta_j^{1*} = \tau^*_{j|i}$.

of the panel. The time period between successive panel waves was about eight to ten weeks.

For one of the items, the sample was split into those with low education (485 parents) and those with high education (490 parents). The question was, "In the past week did any of your children come to you with a problem that was bothering them?" This question measured the regular occurrence of this kind of communication between parent and child. However, what was to be considered a "problem" was not very well defined; therefore, respondents may not have known which answer to choose, resulting in too much turnover. Moreover, one week is not a very long period for measuring communication on serious problems. (On the other hand, memory effects would have increased for a longer period.) Even if this kind of communication did occur regularly, it may not have occurred in the week before one of the interviews, resulting in random turnover for those who were in this situation.

These data were selected as an example for several reasons. First, and most important, they were published in Wiggins ([1955] 1973), which was the first work on MMLC models. Second, the interested reader can re-estimate the results within a few minutes (with PANMARK). Third, many special cases of MMLC models can be demonstrated with these data. But this is also a drawback, because three-wave data are not informative enough to allow us to compare the fit of all models. Other applications on four- and five-wave data (Poulsen 1982; Langeheine and Van de Pol 1990; Van de Pol and Langeheine 1989) show, however, that some MMLC models can very well be rejected.

5.1. *Selecting a Model*

The arguments on response uncertainty and on the short reference period point in the direction of a model for "noisy" data: i.e., a latent Markov model or a model with random turnover. Nevertheless, we tested several other models, first for the $2 \times 2 \times 2$ table of the whole sample, without splitting the sample for education (see Table 2).

Table 2 gives -2 times the loglikelihood ratio (LR) of the present model compared with the saturated model. This LR statistic is asymptotically χ^2 distributed, i.e., for large random samples. Clearly this sample cannot be considered a random sample from a well-defined population. Nevertheless, LR tests are useful in model selection. This sample is just one of the samples that could have resulted from applying the present selection procedure. The LR can be used to test whether a model holds for the superpopulation of samples that could be obtained by repeating this procedure an infinite number of times. Table 2 shows, however, that many models fit these data (very) well, mostly because it has only three waves. Therefore, model selection cannot be based on testing alone.

The latent class model with two latent classes, which says that on a latent level no change has occurred, does not fit the data unless response probabilities are allowed to vary (strongly) over occasions. However, there is no reason why reliabilities should vary as dramatically as they do. Therefore, this model is rejected.

A first-order Markov chain, which says that all observed change is real and that all parents have the same transition probabilities independent of their past, is also rejected because it fails to fit the data.

A second-order Markov model, which relaxes the assumption of independence of the past, cannot be tested with only three panel waves. Because there is only one transition period, which can be considered conditional on the previous measurement, stationarity cannot be tested either. It nevertheless gives a valid description of the data, stating that parents tend to return at occasion 3 to their occasion 1 answer if they had turned to another answer at occasion 2. We feel, however, that returning to previous answers is not the main issue for these data. The second-order Markov model is more appropriate for data on an advertising campaign or an election campaign.

The latent Markov model also gives a valid description of the data. As in the first-order Markov chain, the assumption of stationarity (time-homogeneous transition probabilities) can be tested. The data do not show that change between occasion 1 and occasion 2 was different from change between occasion 2 and occasion 3.

Wiggins (1973) gave a more parsimonious version of the latent Markov model for these data. He argued that on a latent level, the ENABLE program could only increase contact between parent and child: The transition from "yes" to "no" should have probability 0, $\tau_{1|2} = 0$, $\tau_{2|2} = 1$. Furthermore, he assumed that there would be no difference between latent classes 1 and 2 in the probability of a correct response, $\rho_{1|1} = \rho_{2|2}$ and therefore also $\rho_{2|1} = \rho_{1|2}$. These assumptions do not result in a significant decrease in fit.

The mover-stayer model, which says that some people are susceptible for the ENABLE program and others are not, does not fit the data well. This is also true for the black-and-white model (IpS or RrS), which is a special case of the mover-stayer model. Under the mover-stayer model, only 17 percent of the sample can be considered stayers, i.e., people who took part in the program and whose experiences remained unaltered. For the other 83 percent, one Markov chain is not the best way to describe their turnover patterns.

The good fit of Wiggins's latent Markov model confirms our expectation that the data are noisy in some way. Another way to formulate a noisy MMLC model is to assume that part of the sample's responses were independent of their past responses: i.e., their responses can be considered the result of flipping a (biased) coin (equation (4), an independence chain). In the mover-independence model, the stayer chain from the mover-stayer model has been replaced with an independence chain (see Table 2). The difference between the latent Markov model and the mover-independence model is that the latent Markov model assumes that all repondents are equally responsible for noise in the data, whereas the mover-independence model splits the sample into two distinct parts: One part is blamed for all the noise in the data, and the other part goes free.

The mover-independence model is a good way to represent these data. The mover class and independence class are about equiprobable: 49.5 percent and 50.5 percent. (The test for nonstationarity [Table 2] involves only a nonstationary mover chain. If the independence chain had also been defined as nonstationary, the model would not be identified.) Most importantly, for these data the results from the mover-independence model can be well interpreted. The class of parents who displayed independent responses may consist of parents whose children came to them with problems, but not very often or at irregular intervals. At some occasions the problem was told before the reference period, and at other occasions it happened in the reference period.

The probability that this happens in the reference period does not necessarily have to be 0.5. Therefore, we prefer the mover-independence model above the mover-random-response model (with δ^* and τ^* probabilities fixed on 0.5 for the random response chain), although the probabilities in the independence chain are all estimated to be about 0.5.

5.2. *Simultaneous Analysis of Low-Education and High-Education Subsamples*

In this section, we discuss whether there are any differences, in terms of parameters, between low-education parents and high-education parents. MMLC models that incorporate both change and noise turned out to be both plausible and well fitting for the whole superpopulation when the table was collapsed over education. However, restrictions that seem acceptable for the whole population may turn out to be untenable when subpopulations are considered. Actually, a well-fitting model for the whole population may be a badly fitting model for the subpopulations, and vice versa. Nevertheless, a preselection of models by analysis on the whole population seems sensible; otherwise too many models have to be considered.

Table 3 shows the fit of the models chosen on the basis of the previous subsection. A model that relaxes Wiggins's restriction on response probabilities but maintains the restriction $\tau_{1|2h} = 0$ is also considered. Observed and estimated frequencies are given in Appendix B.

First, we assumed that all parameters for low and high education were equal (+++ in Table 3). Then, we relaxed all equality restrictions, one by one, to determine which significantly deteriorated the fit. For all models, at least one equality restriction should be dropped. For each model, we relaxed the restriction that most increased the fit in terms of an LR test. Next, we tested whether relaxing one of the remaining two equality restrictions would improve the fit. When an equality restriction involved more than one degree of freedom, we also tested whether it could be partially dropped (as is the case with the transition probabilities of the mover-random-response model). In Table 3, the fit of each final specification is mentioned after the +++ model. As a benchmark, the fit of the model with no equality restrictions across subpopulations (−−−) is also given. For a given model, the fit cannot surpass that value.

Each +++ model has an acceptable fit at the 0.05 level (but not much lower). It might be argued that to search for a model with a better fit would be to capitalize on chance. However, the fit may be increased significantly by relaxing one or two equality constraints. Moreover, for purposes of interpretation, it is attractive to choose a model with parameters that depend on education.

TABLE 3
The Fit of Selected MMLC Models, by Education

Model	Chain Proportions (π)	Initial Probabilities (δ, δ^*)	Response Probabilities (ρ)	Transition Probabilities (τ, τ^*)	LR	*df*
	Equality Restrictions Across Subpopulations[a]					
Latent Markov		+	+	+	15.8	9
Latent Markov		+	–	+	2.9	7
Latent Markov		–	–	–	2.1	4
Latent Markov, $\tau_{1\mid 2h} = 0$		+	+	+	17.4	10
Latent Markov, $\tau_{1\mid 2h} = 0$		+	–	+	4.6	8
Latent Markov, $\tau_{1\mid 2h} = 0$		–	–	–	3.7	6
Wiggins		+	+	+	17.5	11
Wiggins		–	+	+	10.8	10
Wiggins		–	–	–	8.0	8
Mover-independence	+	+		+	15.0	9
Mover-independence	+	–		+/–	2.6	7
Mover-independence	–	–		–	1.3	4
Mover-random-response	+	+		+	15.0	10
Mover-random-response	+	–		+/–	2.6	8
Mover-random-response	–	–		–	1.7	6

[a]"+" = equal, "–" = not equal, "+/–" = only $\tau^*_{1\mid 111} \neq \tau^*_{1\mid 112}$ and $\tau^*_{2\mid 111} \neq \tau^*_{2\mid 112}$.

The mover-independence and mover-random-response models show that high-education parents are more likely than low-education parents to talk with their children about their children's problems at occasion 1, at least in the mover chain ($\delta^{1*}_{i|1h}$, Table 5). On a latent level ($\delta^{1}_{a|h}$), the same result is found only when response probabilities are assumed to be the same for low-education and high-education parents, as in Wiggins's model. Results from the latent Markov model suggest, however, that Wiggins's restriction on response probabilities, $\rho_{1|1h} = \rho_{2|2h}$ and $\rho_{2|1h} = \rho_{1|2h}$, might be relaxed. Therefore, we evaluated a latent Markov model that maintains only Wiggins's other restriction, $\tau_{1|2h} = 0$. The LR for the $--+$ specification of Wiggins's model, conditional on the same ($--+$) specification of the latent Markov model with only the $\tau_{1|2h} = 0$ restriction, is 6.66 with 2 degrees of freedom (not in a table), which corresponds in the χ^2 distribution to a probability of only 3.6 percent that Wiggins's model holds, given the less restrictive model. Parameter estimates of the latent Markov model with only the $\tau_{1|2h} = 0$ restriction ($+--+$ specification) are given in Table 4.

According to this model, which fits the data well, children of low-education parents are no less likely to go to their parents with problems than children of high-education parents. The apparent difference on a manifest level can be attributed to a stronger tendency of low-education parents to say "no" and a stronger tendency of high-education parents to say "yes." Low-education parents and high-education parents do not differ in the average reliability of their answers, as the final specification of Wiggins's model ($-++$) shows.

Consistent with this finding is that the mover-independence model does not find a difference in the percentage random response between low-education parents and high-education parents. For the rest, however, it offers another interpretation of the differences in response patterns between subpopulations (Table 5). First, low-education parents have less contact with their children about their children's problems than high-education parents. Second, the ENABLE program increases contact with children to a lesser extent for the low-education parents than for the high-education parents. It is interesting to note that Wiggins's assumption that contact cannot decrease, $\tau_{1|2h} = 0$, is the ML estimate for the mover chain for this transition probability.

TABLE 4

Estimated Parameter Values for the Latent Markov Model with $\tau_{1|2h} = 0$, for Low-Education and High-Education Parents
(Standard Errors in Parentheses)

Subpopulation, Class	Sub-population Proportion (γ)	Class Proportion Occasion 1 (δ)	Probability (ρ) of Responding: "No"	Probability (ρ) of Responding: "Yes"	Transition Probability (τ), from t to $t+1$: Class 1	Transition Probability (τ), from t to $t+1$: Class 2
Low-education	0.497 (0.016)					
Class 1 (no)		0.632 (0.042)	0.853 (0.030)	0.147 (0.030)	0.772 (0.025)	0.228 (0.025)
Class 2 (yes)		0.368 (0.042)	0.210 (0.032)	0.790 (0.032)	0	1
High-education	0.503 (0.016)					
Class 1 (no)		0.632 (0.042)	0.788 (0.032)	0.212 (0.032)	0.772 (0.025)	0.228 (0.25)
Class 2 (yes)		0.368 (0.042)	0.120 (0.029)	0.880 (0.029)	0	1

Note: Parameter values of 0 and 1 were fixed.

Dropping the restriction of equal transition probabilities from the latent Markov model with $\tau_{1|2h} = 0$ will not increase the fit significantly. Table 3 shows that the LR value of the $+-+$ specification of this model differs by only 0.9 from the LR value of the $---$ specification, much less than 3.8, which is the 0.05 percentage point of a χ^2 distribution with one degree of freedom. However, when estimates of transition probabilities for a $+--$ specification are compared, low-education parents are also estimated to gain less contact with their children than high-education parents. The probability of remaining in latent class 1 (associated with a "no" response) is estimated to be 0.786(0.036) for low education and 0.759(0.035) for high education.

The nonsignificance of the difference between these parameters can be tested by comparing the LR value of nested models and by looking at the covariances of the parameters. The standard deviation of the difference between $\hat{\tau}_{1|11}$ and $\hat{\tau}_{1|12}$ can be obtained by using

$$\sigma(\hat{\tau}_{1|11} - \hat{\tau}_{1|12}) = [\sigma^2(\hat{\tau}_{1|11}) + \sigma^2(\hat{\tau}_{1|12}) - 2\sigma(\hat{\tau}_{1|11}, \hat{\tau}_{1|12})]^{\frac{1}{2}}.$$

The correlation between these parameters is estimated at 0.04. Hence, the covariance is 0.0000504, and the standard deviation of the difference is 0.049, which is larger than the difference between $\hat{\tau}_{1|11}$ and $\hat{\tau}_{1|12}$, divided by 1.96. This way of testing restrictions may be quicker than fitting another model.

The restricted latent Markov model ($\tau_{1|2h}$) and the mover-independence model point to distinct differences between low-education and high-education parents. The restricted latent Markov model states that low-education parents had a higher tendency to indicate that their children did not come to them with problems than high-education parents and that high-education parents had a higher tendency to indicate that their children did come to them. The mover-random-response model, on the other hand, states that low-education parents had less contact with their children about their children's problems and that they were less influenced by the ENABLE program. The latter finding—less increase in contact with children among low-education parents—is not contradicted by the latent Markov model. On an empirical basis, however, we cannot choose between the two models and their interpretations. Both

TABLE 5
Estimated Parameter Values for the Mover-Independence Model, for Low-Education and High-Education Parents
(Standard Errors in Parentheses)

Subpopulation, Chain, Category	Subpopulation Proportion (γ)	Chain Proportion (π^*)	Category Proportion Occasion 1 (δ^*)	Transition Probability (τ^*)[a] From t to $t+1$ "No"	"Yes"	From $t = 1$ to $t = 3$ "No"	"Yes"
Low-Education	0.497 (0.016)						
Mover chain		0.494 (0.028)					
"No" response			0.720 (0.042)	0.758 (0.034)	0.242 (0.034)	0.574	0.426
"Yes" response			0.280 (0.042)	0.000	1.000	0.000	1.000

Independence		0.506 (0.028)					
"No" response			0.505 (0.030)	0.505 (0.030)	0.495 (0.030)	0.505	0.495
"Yes" response			0.495 (0.030)	0.505 (0.030)	0.495 (0.030)	0.505	0.495
High-education	0.503 (0.016)						
Mover chain		0.494 (0.028)					
"No" response			0.577 (0.044)	0.659 (0.444)	0.341 (0.444)	0.434	0.566
"Yes" response			0.423 (0.444)	0.000	1.000	0.000	1.000
Independence		0.506 (0.028)					
"No" response			0.505 (0.030)	0.505 (0.030)	0.495 (0.030)	0.505	0.495
"Yes" response			0.495 (0.030)	0.505 (0.030)	0.495 (0.030)	0.505	0.495

Note: Parameter values of 0.000 and 1.000 were not fixed but estimated.
[a]Only the $\tau^{*}_{1|11h}$ and $\tau^{*}_{2|11h}$ were not set equal for low and high education.

models give a valid description of the data for this sample size and this number of panel waves.

6. DISCUSSION

Mixed Markov latent class analysis, which was developed by Wiggins (1973) and Poulsen (1982), provides a wide range of latent class models for (panel) data on a characteristic with a limited number of states. These models can describe turnover between "snapshots" from two or more panel waves. This paper has given an overview of these models and has described how an explanatory variable with a limited number of categories can be included in the model, also for only one wave. Differences between these categories in chain proportions, initial probabilities, response probabilities, or transition probabilities can be valuable information.

In the last paragraph of the previous section, two competing interpretations were given for the findings of that section. MaCurdy (1982) faced a similar problem when comparing several ARMA specifications for the error term in a covariance structure model for ten-wave panel data. Some models have similar implications for the expected data, and some data are not informative enough to distinguish between alternatives (too few waves, too few respondents, or too little turnover). Tests on the fit of a model will be more powerful if information on more than three waves is available and if sample size is large enough. Then, more realistic models can also be identified: i.e., mixed Markov models with three chains (Van de Pol and Langeheine 1989) or latent Markov models with more latent classes than observed categories, like the latent mixed Markov model (Langeheine and Van de Pol 1990). Of course, such extended models will be identifiable only if enough turnover is observed. Increasing the observation frequency will yield more information only if the T-way cross-table does not have (many) empty cells.

A model can be selected on the basis of a theory, if provided. However, there is a risk that a good-fitting model for the data will be rejected because it does not comply with theory. On the other hand, a fully empirical approach, comparing the fit of all candidate models using a loglikelihood ratio test, should not be blind to the meaning of the models. In the example in section 5, there was little or no theory, but there was an abundance of MMLC models, especially because so many combinations of equality restrictions between subpopulations could be conceived. Then, a preselection of models can be made by testing several MMLC models on the collapsed table, without the subpopulation variable. One can, however, never be certain that a model that has to be rejected for the collapsed table will not fit for separate subpopulations.

For the selection of appropriate equality constraints across subpopulations for a given model, the following procedure was proposed. First, two extreme specifications are fit, one with all parameters equal across subpopulations, the other with all parameters different between subpopulations. This gives a benchmark for the increase in fit that can be attained by relaxing equality constraints. (If a weakly identified model or a degenerated solution for a subpopulation is found, some restrictions will certainly have to be made.)

Next, the best fitting specification is sought by relaxing the equality restrictions on π, δ, ρ, and τ parameters in a systematic, hierarchical way. The restrictions are removed one by one (sets of parameters, i.e., the set of all τ's, etc.), and the improvement in fit is evaluated with an LR test. If no significant improvement can be obtained, the procedure can be refined by looking at individual parameters instead of sets of parameters. If still no significant improvement is obtained, the procedure stops. If, however, the fit is increased significantly, the most restrictive equality constraint is relaxed, and conditional on this assumption, the procedure is repeated and the next best increase in fit is sought.

Another guideline for improving the fit of a model is the size of first-order derivatives of the loglikelihood with respect to each fixed parameter. In these MMLC models, however, some parameters are restricted in more than one way. The response probabilities, ρ, and the transition probabilities, τ, may be subject to both time-homogeneity assumptions and group-homogeneity assumptions. This makes it difficult, though not impossible, to see which assumption is most strongly violated. For LISREL models, first-order derivatives are no longer considered the best guideline in model selection. Modification indices (Saris, Satorra, and Sörbom 1987) are better measures for the stress of restrictions, because they are scaled on the most relevant dimension: expected improvement in LR fit.

MMLC models describe turnover patterns between occasions, which is not the same as the process that gives rise to these patterns. Continuous time models try to do so by making an assumption about the change in the rate of change. Moreover, continuous-time parameters do not depend on the length of the period between panel waves, whereas discrete-time parameters do. Coleman (1964) introduced a continuous-time version of the latent Markov model for a time-independent rate of change and no heterogeneity in the rate of change. However, estimation of continuous-time parameters from panel data can be problematic. Singer and Spilerman (1976) showed that in some cases, observed transition probabilities are inconsistent with any set of Markovian transition rates. Moreover, Singer and Spilerman (1974), among others, modeled heterogeneity by a nuisance term (or error term), which is assumed to have a distinct theoretical distribution. We feel, however, that panel data are not informative enough to test such distributional assumptions. Davies and Crouchley (1986), although familiar with continuous-time models with a nuisance term, also chose a discrete-time mixed (Markov) model, calling it the nonparametric model. MMLC models have the advantage that no restrictive distributional assumption has to be made.

Of course, MMLC models are a simplification of reality. However, given the limitations of panel data, MMLC models may give interesting insights. Initial probabilities and tables of transition probabilities are relatively easy to understand, even for those who are not familiar with the details of the method. Therefore, it seems worthwhile to try several other kinds of applications. For example, multivariate MMLC analysis of consecutive measurements of the cross-table of a cause variable and an effect variable looks promising.

APPENDIX A: SECOND-ORDER MARKOV CHAIN ANALYSIS WITH A PROGRAM FOR FIRST-ORDER MARKOV CHAINS

PANMARK requires the cross-table of consecutive measurements as input, without indices. The frequencies should be sorted

TABLE A-1
Frequency Input for Second-Order Markov Chain Analysis of a Dichotomy in the Four-Wave Case

f_{1111}	f_{1112}	0	0	0	0	f_{1121}	f_{1122}	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	f_{1211}	f_{1212}	0	0	0	0	f_{1221}	f_{1222}
f_{2111}	f_{2112}	0	0	0	0	f_{2121}	f_{2122}	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	f_{2211}	f_{2212}	0	0	0	0	f_{2221}	f_{2222}

TABLE A-2
Structural Zeroes in the Transition Matrix of xx^1 and xx^2, Indexed by f, g, with Transition Probabilities $\tau_{k|ij}^{3\,12*}$ in Terms of x^1, x^2, and x^3, Indexed by i, j, k

			$g = 1$ $j = 1$	$g = 2$ $j = 1$	$g = 3$ $j = 2$	$g = 4$ $j = 2$
			$k = 1$	$k = 2$	$k = 1$	$k = 2$
$f = 1$	$i = 1$	$j = 1$	$\tau_{1\|11}$	$\tau_{2\|11}$	0	0
$f = 2$	$i = 1$	$j = 2$	0	0	$\tau_{1\|12}$	$\tau_{2\|12}$
$f = 3$	$i = 2$	$j = 1$	$\tau_{1\|21}$	$\tau_{2\|21}$	0	0
$f = 4$	$i = 2$	$j = 2$	0	0	$\tau_{1\|22}$	$\tau_{2\|22}$

APPENDIX B SIMULTANEOUS ANALYSIS OF LOW-EDUCATION AND HIGH-EDUCATION PARENTS: OBSERVED AND EXPECTED FREQUENCIES FOR THE FINAL SPECIFICATION

	Score at Wave				Expected Frequency				
Education	1	2	3	Observed Frequency	Latent Markov Model $+ - +$	Latent Markov Model $\tau_{1\|2} = 0$ $+ - +$	Wiggins's Model $- + +$	Mover-Independence Model $+ - +/-$	Mover-Random-Response Model $+ - +/-$
Low	No	No	No	129	125.9	126.2	120.2	129.0	130.9
Low	No	No	Yes	57	66.9	66.7	68.4	61.6	62.3
Low	No	Yes	No	35	35.5	37.1	36.9	30.8	30.7
Low	No	Yes	Yes	74	70.4	69.3	72.7	71.8	72.3
Low	Yes	No	No	30	29.2	27.6	28.5	31.3	30.7
Low	Yes	No	Yes	33	30.6	33.8	31.9	31.3	30.7
Low	Yes	Yes	No	32	31.2	28.7	25.4	33.9	30.7
Low	Yes	Yes	Yes	95	95.3	95.6	100.9	95.2	96.8
High	No	No	No	94	95.3	95.6	102.5	93.7	92.6
High	No	No	Yes	65	62.6	62.3	60.0	63.9	62.4
High	No	Yes	No	29	31.6	33.6	33.3	31.2	31.0
High	No	Yes	Yes	79	75.7	74.4	71.0	79.1	78.5
High	Yes	No	No	32	29.5	27.9	26.1	31.2	31.0
High	Yes	No	Yes	30	29.4	32.9	36.3	28.3	31.0
High	Yes	Yes	No	26	28.1	25.2	30.7	28.3	31.0
High	Yes	Yes	Yes	135	137.9	138.0	130.2	134.3	132.6

Source: Wiggins 1973, pp. 163–64.
Note: For explication of models, see section 5.2 and Table 3.

by their indices, with the index for the last occasion changing fastest. For a three-wave dataset on a dichotomy, with frequencies f_{ijk}, the data for a normal MMLC model should be input in the sequence $f_{111}f_{112}f_{121}f_{122}f_{211}f_{212}f_{221}f_{222}$. To fit a manifest second-order Markov chain to these data, the frequencies should be input in the sequence $f_{111}f_{112}\ 0\ 0\ 0\ 0\ f_{121}f_{122}f_{211}f_{212}\ 0\ 0\ 0\ 0\ f_{221}f_{222}$. The sequence of input frequencies for a dichotomy in the four-wave case are given in Table A-1.

The transition probabilities of a single Markov chain model (equation (3)) for the *xx* variables will be partially zero (Table A-2). Table A-2 also shows where the second-order probabilities of transition from response *i,j* at occasions 1 and 2 to response *k* at occasion 3, $\tau_{k|ij}^{3\,12^*}$, appear in the transition table of xx^1 and xx^2. The simplifications of the second-order Markov chain that were proposed by Logan (1981) and Raftery (1985) are not incorporated in this approach.

REFERENCES

Abdel-Aty, H. A. H. 1989. "Latent Structure Models for Analyzing Processes of Change: A Latent Class Markov Chain." Unpublished manuscript.

Anderson, T. W. 1954. "Probability Models for Analyzing Time Changes in Attitudes." Pp. 17–66 in *Mathematical Thinking in the Social Sciences*, edited by P. F. Lazarsfeld. Glencoe, IL: Free Press.

Anderson, T. W., and L. A. Goodman. 1957. "Statistical Inference About Markov Chains." *Annals of Mathematical Statistics* 28: 89–110.

Bishop, Y. M. M., S. E. Fienberg, and P. W. Holland. 1975. *Discrete Multivariate Analysis*. Cambridge, MA: MIT Press.

Blumen, I., M. Kogan, and P. J. McCarthy. 1966. "Probability Models for Mobility." Pp. 318–34 in *Readings in Mathematical Social Science*, edited by P. F. Lazarsfeld and N. W. Henry. Cambridge, MA: MIT Press.

Brody, C. J. 1986. "Things are Rarely Black and White: Admitting Gray into the Converse Model of Attitude Stability." *American Journal of Sociology* 92: 657–77.

Bye, B. V., and E. S. Schechter. 1986. "A Latent Markov Model Approach to the Estimation of Response Errors in Multiwave Panel Data." *Journal of the American Statistical Association* 81: 375–80.

Clogg, C. C., S. R. Eliason, and J. Grego. 1989. "Models for the Analysis of Change in Discrete Variables." In *New Statistical Models for the Analysis of Longitudinal Data*, edited by A. von Eye. New York: Academic Press.

Clogg, C. C., and L. A. Goodman. 1984. "Latent Structure Analysis of a Set of Multidimensional Contingency Tables." *Journal of the American Statistical Association* 79: 762–71.

———. 1985. "Simultaneous Latent Structure Analysis in Several

Groups." Pp. 81–110 in *Sociological Methodology 1985*, edited by N. B. Tuma. San Francisco: Jossey-Bass.

———. 1986. "On Scaling Models Applied to Data from Several Groups." *Psychometrika* 51: 123–35.

Coleman, J. S. 1964. *Models of Change and Response Uncertainty*. Englewood Cliffs: Prentice-Hall.

Converse, P. E. 1964. "The Nature of Belief Systems in Mass Publics." Pp. 206–61 in *Ideology and Discontent*, edited by D. E. Apter. New York: Free Press.

Davies, R. B., and R. Crouchley. 1986. "The Mover-Stayer Model: Requiescat in Pace." *Sociological Methods and Research* 14: 356–80.

Dayton, C. M., and G. B. Macready. 1983. "Latent Structure Analysis of Repeated Classifications with Dichotomous Data." *British Journal of Mathematical and Statistical Psychology* 36: 189–201.

De Leeuw, J., P. G. M. van der Heijden, and P. Verboon. 1990. "A Latent Time-Budget Model." *Statistica Neerlandica* 44, in press.

Dempster, A. P., N. M. Laird, and D. B. Rubin. 1977. "Maximum Likelihood Estimation from Incomplete Data via the EM Algorithm." *Journal of the Royal Statistical Society*, ser. B, 39: 1–38.

Duncan, O. D. 1985. "New Light on the 16-Fold Table." *American Journal of Sociology* 91: 88–128.

Duncan, O. D., M. Stenbeck, and C. J. Brody. 1988. "Discovering Heterogeneity: Continuous Versus Discrete Latent Variables." *American Journal of Sociology* 93: 1305–21.

Fuchs, C., and J. B. Greenhouse. 1988. "The EM Algorithm for Maximum Likelihood Estimation in the Mover-Stayer Model." *Biometrics* 44: 605–13.

Goodman, L. A. 1974. "Exploratory Latent Structure Analysis Using Both Identifiable and Unidentifiable Models." *Biometrika* 61: 215–31.

Hagenaars, J. A. 1988. "Latent Structure Models with Direct Effects Between Indicators." *Sociological Methods and Research* 16: 379–405.

Heckman, J., and B. Singer. 1984. "A Method for Minimizing the Impact of Distribution Assumptions in Econometric Models for Duration Data." *Econometrica* 52: 271–320.

Jöreskog, K. G. 1971. "Simultaneous Factor Analysis in Several Groups." *Psychometrika* 36: 409–26.

Langeheine, R. 1988. "New Developments in Latent Class Theory." Pp. 77–108 in *Latent Trait and Latent Class Models*, edited by R. Langeheine and J. Rost. New York: Plenum.

Langeheine, R., and F. van de Pol. 1989. "Black-and-White, Black-Gray-White or True Change? A Reexamination of the Converse Model of Attitude Stability." Unpublished manuscript.

———. 1990. "A Unifying Framework for Markov Modeling in Discrete Space and Discrete Time." *Sociological Methods and Research* 18: 416–41.

Lazarsfeld, P. F. 1978. "Some Episodes in the History of Panel Analysis." Pp. 249–65 in *Longitudinal Research on Drug Use*, edited by D. B. Kandel. New York: Halsted Press.

Lazarsfeld, P. F., and N. W. Henry. 1968. *Latent Structure Analysis*. Boston:

Houghton-Mifflin.
Logan, J. A. 1981. "A Structural Model of the Higher-Order Markov Process Incorporating Reversion Effects." *Journal of Mathematical Sociology* 8: 75–89.
MaCurdy, T. E. 1982. "The Use of Time Series Processes to Model the Error Structure of Earnings in Longitudinal Data Analysis." *Journal of Econometrics* 18: 83–114.
Markus, G. B. 1979. "Analyzing Panel Data." Sage University Papers No. 18. London: Sage.
Morgan, T. M., C. S. Aneshensel, and V. A. Clark. 1983. "Parameter Estimation for Mover-Stayer Models, Analyzing Depression Over Time." *Sociological Methods and Research* 11: 345–66.
Plewis, I. 1985. *Analyzing Change: Measurement and Explanation Using Longitudinal Data*. New York: Wiley.
Poulsen, C. S. 1982. "Latent Structure Analysis with Choice Modeling Applications." Ph.D. diss., Wharton School, University of Pennsylvania.
Raftery, A. E. 1985. "A Model for High-Order Markov Chains." *Journal of the Royal Statistical Society*, ser. B, 47: 528–39.
Saris, W. E., A. Satorra, and D. Sörbom. 1987. "The Detection and Correction of Specification Errors in Structural Equation Models." Pp. 105–29 in *Sociological Methodology 1987*, edited by C. C. Clogg. Washington, DC: American Sociological Association.
Singer, B., and S. Spilerman. 1974. "Social Mobility Models for Heterogeneous Populations." Pp. 356–401 in *Sociological Methodology 1973–74*, edited by H. L. Costner. San Francisco: Jossey-Bass.
————. 1976. "The Representation of Social Processes by Markov Models." *American Journal of Sociology* 92: 1–54.
Taylor, M. C. 1983. "The Black-and-White Model of Attitude Stability: A Latent Class Examination of Opinion and Nonopinion in the American Public." *American Journal of Sociology* 89: 373–401.
Tuma, N. B., and M. T. Hannan. 1984. *Social Dynamics: Models and Methods*. New York: Academic Press.
Van de Pol, F., and J. de Leeuw. 1986. "A Latent Markov Model to Correct for Measurement Error." *Sociological Methods and Research* 15: 118–41.
Van de Pol, F., and R. Langeheine. 1989. "Mixed Markov Models, Mover-Stayer Models, and the EM Algorithm, With an Application to Labour Market Data from the Netherlands Socioeconomic Panel." Pp. 485–95 in *Multiway Data Analysis*, edited by R. Coppi and S. Bolasco. Amsterdam: North-Holland.
Van de Pol, F., R. Langeheine, and W. de Jong. 1989. *PANMARK User's Manual, PANel analysis using MARKov chains*. Version 1.5. Voorburg: Netherlands Central Bureau of Statistics.
Van der Heijden, P. G. M., A. Mooijaart, and J. de Leeuw. 1989. "Latent Budget Analysis." Pp. 301–13 in *Statistical Modelling*, edited by A. Decarli, B. Francis, R. Gillcrist, and G. U. H. Seeber. New York: Springer Verlag.
Wiggins, L. M. (1955) 1973. *Panel Analysis, Latent Probability Models for Attitude and Behavior Processes*. Amsterdam: Elsevier.